

WASP-36b

R-2317

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_130313 · created 2026-08-18T00:40:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456364.65682 +/- 0.00054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.015
Transit depth (Rp/Rs)^2	2.0 +/- 0.43 [%]
Orbital Inclination (inc)	88.07 +/- 1.53
Airmass coefficient 1 (a1)	0.817 +/- 0.013
Airmass coefficient 2 (a2)	0.154 +/- 0.037
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	#2 - [473, 93]
Optimal Aperture	3.96
Optimal Annulus	5.79
Transit Duration (day)	0.0837 +/- 0.0016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.015 vs 0.1368 (deviation 0.33σ)
Duration fit vs published	0.0837 d vs 0.0758 d (deviation 4.75σ)
Mid-time O–C	1.3 min
Depth SNR	9.1
Residual scatter	1.6 %

Light curve

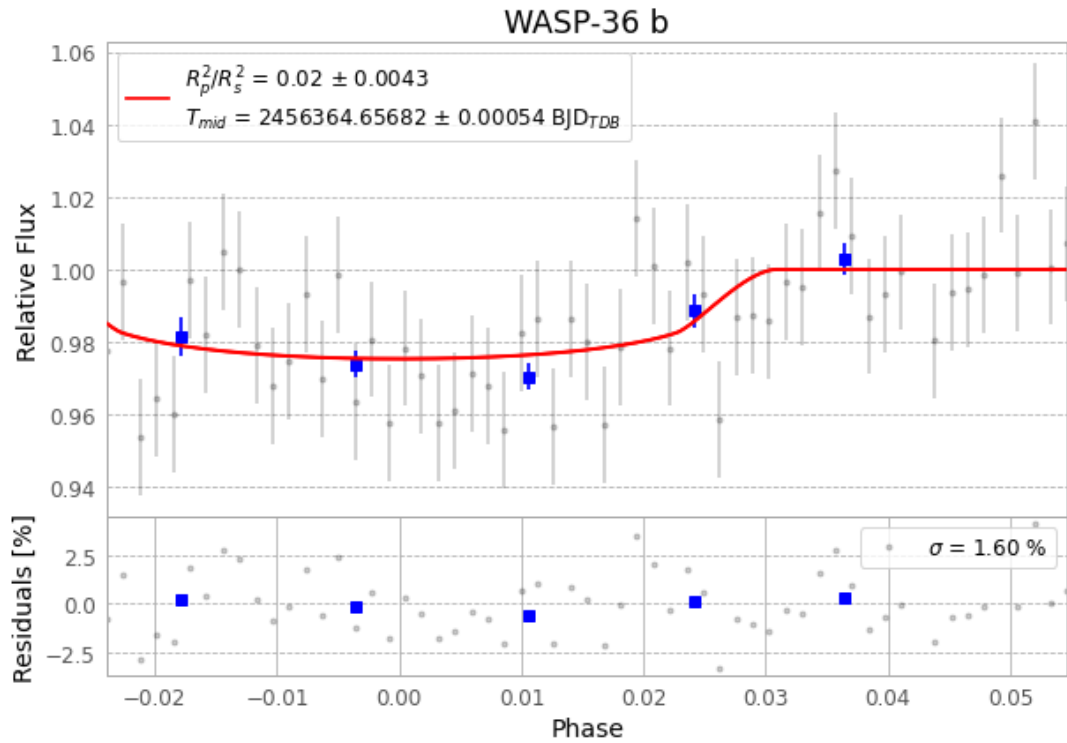


Figure 1 — FinalLightCurve_WASP-36 b_2013-03-12.png (SHA-256 09db9c8f957ef716...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [322, 285], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e56507d0f51ca99c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

59 FITS frames (39 MB), WASP-36130313024512.FITS → WASP-36130313053913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-130313.log (a8b8baa451e6...), FinalLightCurve_WASP-36 b_2013-03-12.png (09db9c8f957e...), FinalLightCurve_WASP-36 b_2013-03-12.csv (654f06d9bc02...)

Compute resources

Wall time 164.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 00:40Z.